

Cubes — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

- Q1** (b) 7. Units digit 3, and $3^3 = 27$ ends in 7.
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- Q2** (b) 125. $125 = 5^3$; the others are not whole numbers cubed.
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- Q3** 64. $4 \times 4 \times 4 = 64$.
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- Q4** (b) 4. They are 1, 8, 27, 64.
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- Q5** False. A cube's trailing zeros must come in multiples of three.
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- Q6** (a) Both true, R explains A. $1728 = 2^6 \times 3^3 = 12^3$; all primes form triples, which is exactly why it is a cube.
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Short answer

- Q7** (i) 3 (ii) 4. 7^3 ends in 3; $4^3 = 64$ ends in 4.
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- Q8** $27 = 7 + 9 + 11$. (The 3rd group of consecutive odd numbers = 3^3 .)
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- Q9** No. $256 = 2^8$; 8 is not a multiple of 3, so the 2s do not form complete triples.
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Long answer

- Q10** Cube root = 15. $3375 = (3 \times 3 \times 3)(5 \times 5 \times 5)$, so $\sqrt[3]{3375} = 3 \times 5 = 15$.
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- Q11** Multiply by 60. $3600 = 2^4 \times 3^2 \times 5^2$; need $2^2 \times 3 \times 5 = 60$ to complete the triples ($3600 \times 60 = 216000 = 60^3$).
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- Q12** Divide by 4; $\sqrt[3]{64} = 4$. $256 = 2^8$; remove $2^2 \rightarrow 2^6 = 64 = 4^3$.
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Total: 21 marks.